

WASP-2b

R-0261

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_240723 · created 2026-07-17T08:58:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460514.9094 +/- 0.0079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.046
Transit depth (Rp/Rs)^2	2.55 +/- 1.48 [%]
Orbital Inclination (inc)	84.09 +/- 2.85
Airmass coefficient 1 (a1)	1.34 +/- 0.64
Airmass coefficient 2 (a2)	-0.24 +/- 0.35
Scatter in the residuals of the lightcurve fit is	48.17 %
Best Comparison Star	None
Optimal Aperture	2.9
Optimal Annulus	6.61
Transit Duration (day)	0.058 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.046 vs 0.1326 (deviation 0.6σ)
Duration fit vs published	0.058 d vs 0.0737 d (deviation 0.45σ)
Mid-time O–C	13.3 min
Depth SNR	3.5
Residual scatter	48.17 %

Light curve

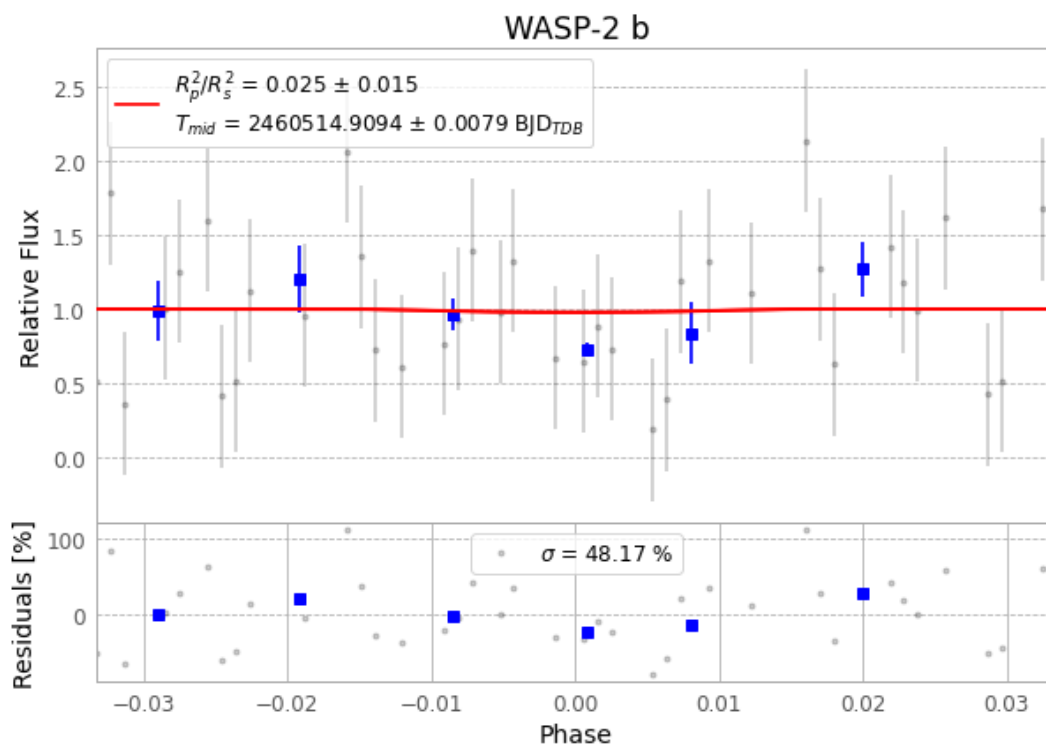


Figure 1 — FinalLightCurve_WASP-2 b_2024-07-23.png (SHA-256 6d7fa5b50a149517...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [540, 306], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e724cc021f11591f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

71 FITS frames (47 MB), WASP-2240723075416.FITS → WASP-2240723112415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-240723.log (584118834b01...), FinalLightCurve_WASP-2 b_2024-07-23.png (6d7fa5b50a14...), FinalLightCurve_WASP-2 b_2024-07-23.csv (b985511e96f1...)

Compute resources

Wall time 210.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 08:58Z.